

Summary

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What we need to know about functions

- State the definition of a function.
- Find the domain and range of a function.
- Distinguish between functions by considering their domains.
- Recognize different representations of the same function.
- Determine where a function is positive or negative.
- Determine algebraically whether a function is even, odd, or neither.
- Use symmetry when graphing even or odd functions.
- Plot basic functions.
- Apply appropriate transformations to graphs of basic functions (vertical and horizontal shifts, vertical and horizontal stretching and reflecting)
- Perform basic operations and compositions on functions.
- Work with piecewise defined functions.
- Determine if a function is one-to-one.
- Define and work with inverse functions.
- Plot inverses of basic functions.
- Find inverse functions (algebraically and graphically).
- Find the largest interval containing a given point where the function is invertible.
- Determine the intervals on which a function has an inverse.
- Relate the domain and range of f and the domain and range of f^{-1} .
- Use a sign-chart to determine where the function is positive/negative.